ICT2142 · Database Systems

Foundations: Relational Data and Keys

Original synthetic reading for demonstration and independent study. It is not an official syllabus, lecture note, or assessment specification.

Learning objectives

After this reading, a learner should be able to explain the central idea, apply it to a small example, and identify at least one assumption or limitation.

1. Main idea

A relational database stores facts in tables whose rows represent records and columns represent attributes. A primary key identifies a row; a foreign key enforces a relationship to another table. Candidate keys and unique constraints protect natural identifiers.

2. Example and reasoning

In a university schema, a course code can be a natural unique key while an internal course_id is a surrogate key. Enrollments reference students and offerings, so both parent records must exist first.

3. A method for practice

1. State the problem and define the terms that affect the answer.
2. Write down the assumptions and information available.
3. Work through a small example and make each step visible.
4. Check the result against a second case or independent explanation.
5. Summarize what the result supports and what it does not establish.

4. Questions for study

- Explain “foundations: relational data and keys” in your own words and give a new example.
- Which assumption matters most in the example, and how would the answer change if it were false?
- Describe one common mistake and a practical way to detect it.
- Write a short checklist another learner could use to review this work.

Reflection

Separate evidence from assumptions. If a conclusion depends on information not provided here, identify what would need to be verified.